LATENT FLOW + ANOVA TOP-256 + PLS-DA

Patient-level spectral classification | fold-specific augmentation and feature selection

01 INPUT & FOLD SPLIT

Original spectra
N = 17 patients

Leave-one-out split
1 patient held out

Real training spectra
SNV normalization

02 CLASS-CONDITIONAL LATENT FLOW

PCA
d = 4

Latent space
standardization

Conditional
flow matching

Inverse PCA
+ SNV

Gaussian source → learned velocity → synthetic latent sample

350 gradient steps • Heun ODE solver (32 steps) • 30 synthetic spectra per class

03 FEATURE SELECTION & CLASSIFIER

Augmented training spectra
SNV-normalized real + 60 synthetic spectra

ANOVA Top-256
Rank each spectral coordinate by class F-statistic

PLS-DA classifier
4 latent components • threshold = 0.5

04 HELD-OUT PATIENT EVALUATION

No synthetic samples in the test set

Held-out original
1 patient per fold

SNV normalization
applied to test spectrum

ANOVA-selected features
apply the 256 training-fold indices

Fitted PLS-DA
predict using trained model

Patient prediction
healthy / diseased

Training-only fitting: PCA, latent flow, ANOVA ranking, and PLS-DA are rebuilt independently for every fold.

Reported metrics use out-of-fold predictions from original patients only.